

Full Stack AI/ML Engineering Bootcamp

বুটক্যাম্প সম্পর্কিত তথ্য

- Complete AI/ML Engineering with Model Deployment.
 - টোটাল ক্লাসঃ ৩৪ টি
 - টোটাল অ্যাসাইনমেন্টঃ ১৪
 - টোটাল প্রোজেক্টঃ ১৪
 - ডিউরেশনঃ ৬-৮ মাস
- Thesis & Research Paper Guideline & Support

ML/AI career paths:

ML + Analytics: Data Scientist, ML Analyst

Academic Research: Research Scientist, PhD path

ML + Software Engineering: ML Engineer, AI Engineer

ভালোভাবে পড়ুন

পুরো বুটক্যাম্পে আমরা প্রত্যেকটা ক্লাসের উপর শিক্ষার্থীদের ক্লাস এটেন্ডেন্স, এসাইনমেন্ট সাবমিশন ট্র্যাকিং এ রাখবো এবং মার্কিং করবো।

10 April

Online Orientation

9:00 PM

Module	Class No	Topic	Date	Time	Mentor
	Class - 1	Introduction - Data Science overview - AI vs ML vs Deep Learning - Supervised vs Unsupervised Learning - Regression, Classification, Clustering - Real-world use cases - Career paths for ML/AI domain (ML+Analytics vs. Academic vs. ML+SWE) - Required tools - Required platforms - Additional materials on Linear Algebra and Statistics			
Module - 01 : Python Fundamentals					
	Class - 2	Python Setup & Basics - Setting up IDE (VS Code, PyCharm, Jupyter) - Python syntax and basic operations - Variables and data types - Input/output operations - Basic arithmetic and string operations			
	Class - 3	Control Flow & Functions - Conditional statements & Loops - Control statements - Functions & Arguments - List - Tuple - Set - Dictionary - When and why to use each structure - List comprehensions			
	Assignment - 01 : Python Problem Solving				

	Class - 4	Advanced Python Basics & File Handling - Lambda functions - Map, Filter, Reduce functions - List, Dictionary, Set comprehensions - File handling (open, read, write, append) - Working with CSV files - JSON file operations - Error handling with try-except - String formatting (f-strings, format()) - Working with modules (import, from) - Command line arguments - Regular expressions basics (re module)			
Assignment - 02 : Advanced Python Practice					
Module - 02 : Data Manipulation & Analysis					
	Class - 5	NumPy Fundamentals - NumPy arrays vs Python lists - Array creation and operations - Indexing and slicing - Mathematical operations - Broadcasting - Random number generation - Array reshaping			
	Class - 6	Pandas for Data Manipulation - Series and DataFrame - Reading/writing data (CSV, Excel, JSON) - Data selection and filtering - Handling missing values - Data aggregation and groupby - Merging and joining datasets - Basic time series operations			
Assignment - 03 : Data Manipulation with Pandas					
	Class - 7	Data Cleaning & Preprocessing - Understanding different data types - Identifying and handling missing values (mean, median, mode, forward/backward fill) - Outlier detection (IQR, Z-score) - Outlier treatment (capping, removal) - Data type conversion - Duplicate removal - String operations for text cleaning			

	Class - 8	Exploratory Data Analysis (EDA) & Visualization - EDA methodology - Statistical summaries (describe, info) - Distribution analysis - Correlation analysis - Matplotlib basics (line, bar, scatter plots) - Seaborn for statistical plots - Plotly for interactive visualizations			
Assignment - 04 : Comprehensive EDA & Visualization					
Module - 03 : Feature Engineering & ML Preparation					
	Class - 9	Feature Engineering & Encoding - Feature scaling (StandardScaler, MinMaxScaler, RobustScaler) - Normalization techniques - Encoding categorical variables (Label, One-Hot, Ordinal) - Feature creation and extraction - Binning and discretization - Handling date-time features			
	Class - 10	Train-Test Split & Data Leakage Prevention - Importance of data splitting - Train-test-validation split - Stratified sampling - Cross-validation (K-Fold, Stratified K-Fold) - Data leakage concepts and prevention - Handling imbalanced datasets (SMOTE, SMOTETomek, class weights) - Pipeline creation for preprocessing			
Assignment - 05 : Feature Engineering & Data Preparation					
Module - 04 : Introduction to Machine Learning					
	Class - 11	ML Fundamentals & First Model - AI vs ML vs Deep Learning - Supervised vs Unsupervised Learning - Regression vs Classification - ML workflow overview - Introduction to scikit-learn - Building first Linear Regression model - Model evaluation basics			

	Class - 12	Model Evaluation & Metrics - Regression metrics (MAE, MSE, RMSE, R^2) - Classification metrics (Accuracy, Precision, Recall, F1) - Confusion matrix - ROC curve and AUC - Overfitting vs Underfitting - Bias-Variance tradeoff - Learning curves			
Assignment - 06 : First ML Model & Evaluation					
Module - 05 : Regression Algorithms					
	Class - 13	Linear & Polynomial Regression - Simple Linear Regression theory - Multiple Linear Regression - Loss functions (MSE, MAE) - Polynomial Regression - When to use polynomial features - Feature importance in regression			
Project 01 : House Price Prediction System					
	Class - 14	Regularization Techniques - Ridge Regression (L2 regularization) - Lasso Regression (L1 regularization) - ElasticNet (L1 + L2) - When to use each technique - Feature selection with Lasso - Hyperparameter tuning for alpha			
	Class - 15	KNN & Distance-Based Learning - K-Nearest Neighbors algorithm - Distance metrics (Euclidean, Manhattan) - Choosing optimal K - KNN for regression and classification - Advantages and limitations - Computational complexity			
Assignment - 07 : Regression Models Comparison					
Module - 06 : Classification Algorithms					

	Class - 16	Logistic Regression - Binary classification - Sigmoid function - Decision boundary - Multiclass classification (One-vs-Rest, Softmax) - Probability predictions - Classification threshold tuning			
Project - 02 : Disease Prediction System					
	Class - 17	Support Vector Machines (SVM) - Margin and hyperplane concepts - Support vectors - Kernel trick (Linear, RBF, Polynomial) - C parameter (regularization) - Gamma parameter - SVM for binary and multiclass			
	Class - 18	Naïve Bayes & Text Classification - Probability fundamentals - Bayes theorem - Naïve Bayes assumptions - Types (Gaussian, Multinomial, Bernoulli) - Text preprocessing (lowercase, remove punctuation, stopwords) - Text vectorization (CountVectorizer, TfidfVectorizer)			
Project - 03 : Email Spam Classifier					
	Class - 19	Decision Trees - Decision tree structure - Information gain and Gini impurity - Tree building process - Pruning techniques - Feature importance - Visualization of decision trees - Hyperparameters (max_depth, min_samples_split)			
Assignment - 08 : Classification Models Comparison					
Module - 07 : Ensemble Learning					

	Class - 20	Ensemble Methods - Bagging & Boosting - Ensemble learning concept - Bagging (Bootstrap Aggregating) - Random Forest algorithm - Boosting concept - Gradient Boosting basics - XGBoost, CatBoost, LightGBM - Feature importance in ensemble methods - When to use which ensemble method			
Project - 04 : Customer Churn Prediction					
	Class - 21	Hyperparameter Tuning & Model Selection - Grid Search - Random Search - Bayesian Optimization concept - Automated tuning with Optuna - Cross-validation with tuning - Model selection strategies - Best practices for hyperparameter tuning			
Assignment - 09 : Ensemble Methods & Hyperparameter Tuning					
Module - 08 : Unsupervised Learning & Time Series					
	Class - 22	Clustering Algorithms - Unsupervised learning overview - K-Means clustering algorithm - Elbow method for optimal K - Silhouette score - Hierarchical clustering - DBSCAN introduction - Cluster evaluation			
Project - 05 : Customer Segmentation System					
	Class - 23	Time Series Forecasting - Time series basics - Components (trend, seasonality, residual) - Stationary vs non-stationary data - ARIMA model - SARIMA for seasonal data - Prophet by Facebook - Evaluation metrics for time series			
Project - 06 : Stock Market Price Forecasting					
Module - 09 : Advanced NLP & Projects					

	Class - 24	Advanced NLP & Sentiment Analysis - Advanced text preprocessing - N-grams - Word embeddings concept - Sentiment analysis techniques - Building an end-to-end NLP pipeline - Model deployment for text classification			
Project - 07 : Fake News Detection System					
	Class - 25	End-to-End ML Project Workflow - Complete ML project lifecycle - Problem definition and scoping - Data collection strategies - Comprehensive EDA - Model selection criteria - Evaluation and comparison - Documentation best practices - Presentation skills			
Project - 08 : Amazon Product Review Sentiment Analysis					
Module - 10 : Model Deployment					
	Class - 26	Web Apps with Streamlit & Gradio - Introduction to Streamlit - Building interactive dashboards - Model integration with Streamlit - Gradio for quick ML interfaces - User input handling - File upload functionality - Deployment on Streamlit Cloud/Hugging Face Spaces - Best practices for UI/UX			
Project - 09 : Multi-Model ML Dashboard					

	Class - 27	FastAPI for ML Deployment - What is FastAPI and why for ML - REST API basics (GET, POST) - Creating ML inference API - Request/response schema with Pydantic - Model serialization (pickle, joblib) - Loading and using saved models - Testing with Swagger UI - Error handling and validation - CORS configuration			
Assignment - 10 : ML Model API Development					
Module - 11 : Deep Learning with PyTorch					
	Class - 28	Neural Networks Fundamentals - What is Deep Learning? - Neuron and Perceptron model - Artificial Neural Networks (ANN) architecture - Forward propagation - Activation functions (Sigmoid, Tanh, ReLU, Leaky ReLU, Softmax) - Loss functions (MSE, Binary Cross-Entropy, Categorical Cross-Entropy) - Backpropagation concept (intuition) - Gradient Descent and learning rate - Optimizers (SGD, Momentum, Adam, RMSprop) - Batch, Mini-batch, and Stochastic gradient descent - Overfitting in neural networks - Regularization techniques (Dropout, L1/L2) Hands-on: - Build perceptron from scratch using NumPy - Visualize activation functions - Understand gradient descent with simple examples			

	Class - 29	PyTorch, Keras Basics & Building Neural Networks - Introduction to PyTorch and Keras - Why PyTorch for Deep Learning? - Tensors and tensor operations - PyTorch vs NumPy - Autograd and automatic differentiation - Building neural networks with nn.Module - Defining layers (Linear, Activation) - Dataset and DataLoader - Training loop structure - Evaluation loop - Saving and loading models - GPU acceleration basics			
	Project - 10 : Image Classification with ANN				
	Class - 30	Convolutional Neural Networks (CNN) - Limitations of ANN for images - Introduction to CNN - Convolution operation - Filters and feature maps - Pooling layers (Max pooling, Average pooling) - CNN architecture (Conv → Pool → FC) - Famous architectures overview (LeNet, AlexNet, VGG) - Transfer Learning concept - Data augmentation techniques - Building a CNN with PyTorch			
	Project - 11 : Advanced Image Classification with CNN				
	Assignment - 11 : CNN Implementation & Comparison				

	Class - 31	Recurrent Neural Networks (RNN) & Sequence Models - Sequential data and its challenges - Introduction to RNN - RNN architecture and working - Vanishing gradient problem - Long Short-Term Memory (LSTM) - Gated Recurrent Unit (GRU) - Bidirectional RNN - Sequence-to-sequence models - Applications: Text generation, time series, translation - Building RNN/LSTM with PyTorch			
	Project - 12 : Text Generation & Sentiment Analysis with LSTM Option A - Text Generation Option B - Multilingual Sentiment Analysis				
	Assignment - 12 : Deep Learning Projects & Documentation				

	Class - 32	<i>Transformers & Attention Mechanisms</i> - Limitations of RNNs and motivation for Transformers - Self-attention mechanism i) Query, Key, Value concepts ii) Scaled dot-product attention iii) Multi-head attention - Transformer architecture i) Encoder-decoder structure ii) Positional encoding iii) Feed-forward networks iv) Layer normalization and residual connections - Pre-trained Transformer models i) BERT (Bidirectional Encoder Representations from Transformers) ii) GPT (Generative Pre-trained Transformer) iii) T5 (Text-to-Text Transfer Transformer) - Transfer learning and fine-tuning - Hugging Face Transformers library - Applications: Text classification, NER, question answering, summarization - Vision Transformers (ViT) - brief introduction			
	<i>Project - 13 : Transformer-based NLP Application</i>				
	<i>Option A - Text Classification with BERT</i> <i>Option B - Question Answering System</i>				
	<i>Assignment - 13 : Transformer Experimentation</i>				
Module - 12 : Thesis, Publication, Documentation & Career Guideline					

	Class - 33	LaTeX, Thesis Writing & Career Guidelines - 1 - Introduction to LaTeX - Why LaTeX for academic writing - Setting up an Overleaf account - Document structure (sections, subsections) - Mathematical equations formatting - Tables and figures - Citations and bibliography (BibTeX) - Cross-referencing - Research paper template - Thesis formatting guidelines - Conference/Journal Publication Guideline Hands-on: - Create a research paper template - Write mathematical equations - Add tables and figures - Manage references - Export to PDF			
	Class - 34	LaTeX, Thesis Writing & Career Guidelines - 2 - ML/AI career paths: i) ML + Analytics: Data Scientist, ML Analyst ii) Academic Research: Research Scientist, PhD path iii) ML + Software Engineering: ML Engineer, AI Engineer - Building professional resume for ML roles - GitHub portfolio optimization - LinkedIn profile optimization - Kaggle competitions strategy - Interview preparation (technical + behavioral) - Networking in ML community - Continuous learning resources - Freelancing opportunities - Building personal brand			
	Assignment - 14 : Documentation & Career Preparation				
	FINAL CAPSTONE PROJECT				

